ECE 3210 hw09

1. Given the differential equation

$$(D^2 + 7D + 12)y(t) = (3D + 9)f(t),$$

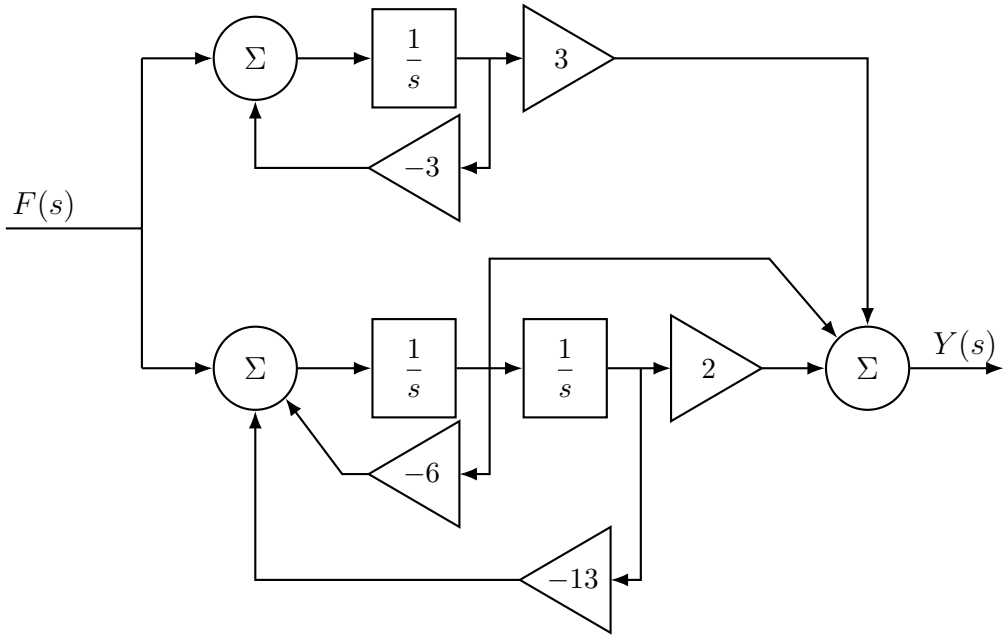
where $y(0^-) = 1$, $y'(0^-) = 2$, and $f(t) = e^{-3t}u(t)$, find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

(b) Find the zero-state response $y_{\text{zs}}(t)$.

(c) Find the total response $y_{\text{tot}}(t)$.

2. Given the parallel system realization below, find the system's impulse response $h(t)$.



3. Given the differential equation

$$(D^2 + 6D + 25)y(t) = (D + 1)f(t),$$

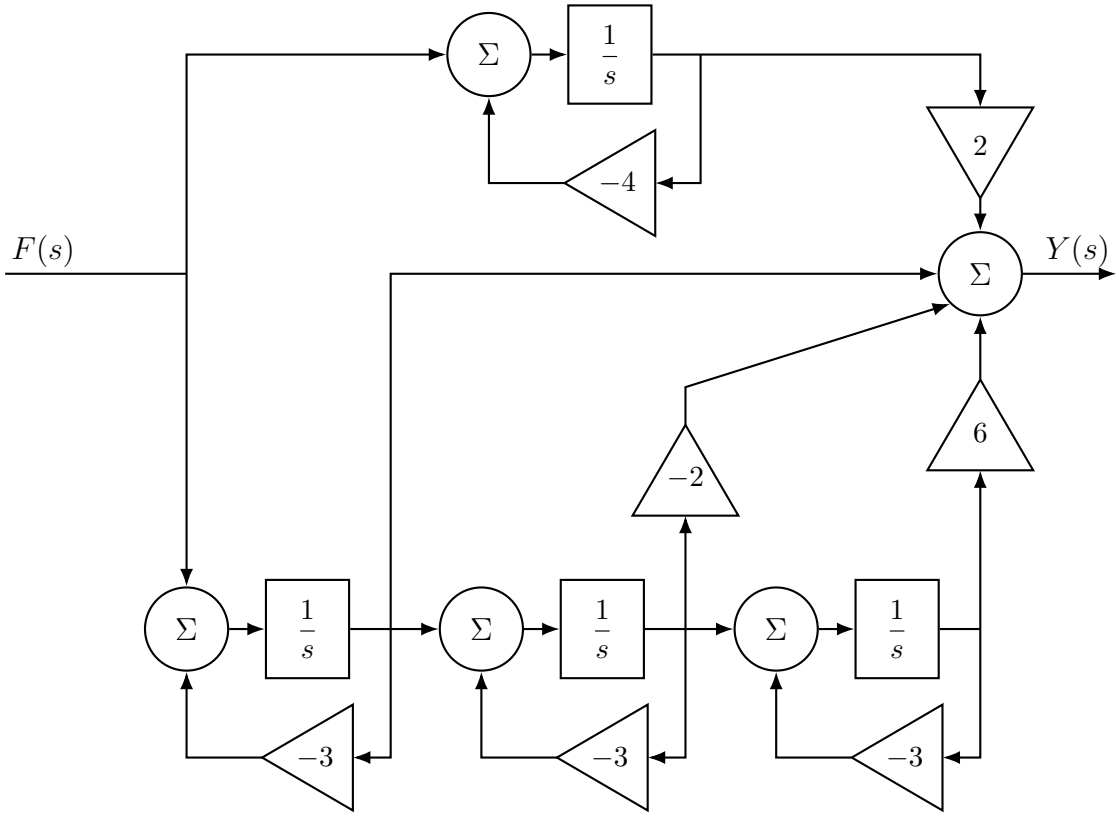
where $y(0^-) = 1$, $y'(0^-) = 2$, and $f(t) = u(t)$, find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

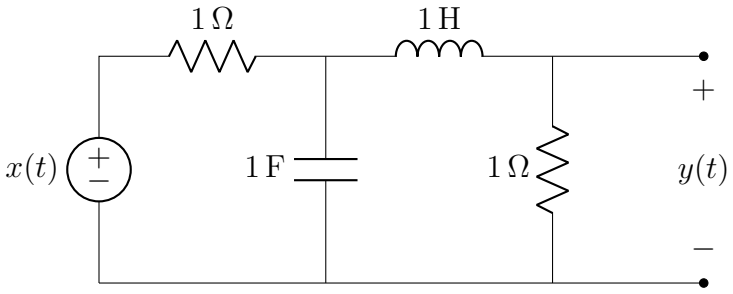
(b) Find the zero-state response $y_{\text{zs}}(t)$.

(c) Find the total response $y_{\text{tot}}(t)$.

4. Given the system realization below, find the system's impulse response $h(t)$.



5. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

(b) Given the input $x(t) = e^{-t}u(t)$, find the zero-state response $y(t)$.

6. Consider the differential equation

$$(D^2 + 4D + 4)y(t) = (2D^2 + 11D + 8)f(t),$$

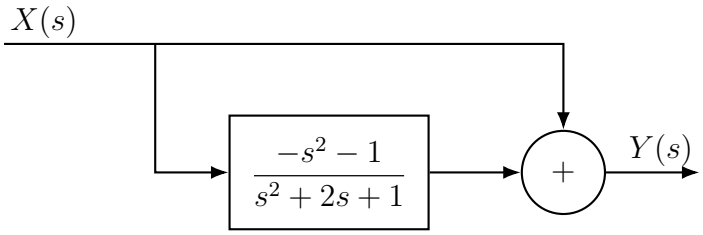
with $y(0^-) = 4$, $y'(0^-) = -6$, and input $f(t) = u(t)$. Find the following responses using the Laplace transform.

(a) Find the zero-input response $y_{\text{zi}}(t)$.

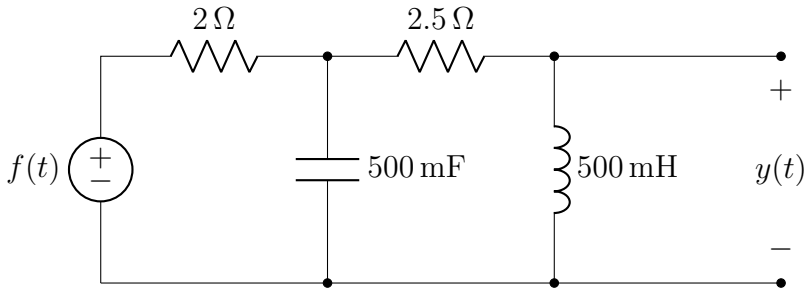
(b) Find the zero-state response $y_{\text{zs}}(t)$.

(c) Find the total response $y_{\text{tot}}(t)$.

7. Consider the block diagram below. Find the impulse response $h(t)$ for this system.



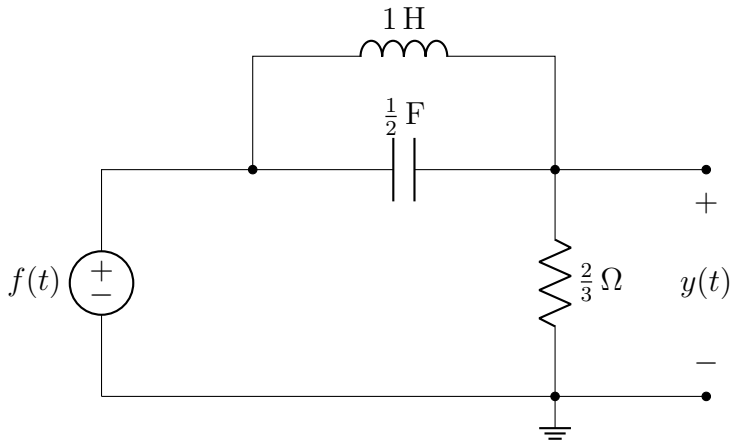
8. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

(b) Given the input $x(t) = u(t)$, find the zero-state response $y(t)$.

9. Consider the circuit below.



(a) Derive the transfer function $H(s)$ for this circuit.

(b) What type of filter is this?

(c) Find the impulse response $h(t)$.

(d) Given the input $f(t) = u(t)$, find the zero-state response $y(t)$.

Answer Key

1(a) $y_{\text{zi}}(t) = \big(6e^{-3t} - 5e^{-4t}\big)u(t)$
1(b) $y_{\text{zs}}(t) = 3\big(e^{-3t} - e^{-4t}\big)u(t)$
1(c) $y_{\text{tot}}(t) = \big(9e^{-3t} - 8e^{-4t}\big)u(t)$
2 $h(t) = e^{-3t}\big(3 + \cos(2t) - \frac{1}{2}\sin(2t)\big)u(t)$
3(a) $y_{\text{zi}}(t) = e^{-3t}\big(\cos(4t) + \frac{5}{4}\sin(4t)\big)u(t)$
3(b) $y_{\text{zs}}(t) = \big(\frac{1}{25} + e^{-3t}(-\frac{1}{25}\cos(4t) + \frac{11}{50}\sin(4t))\big)u(t)$
3(c) $y_{\text{tot}}(t) = \big(\frac{1}{25} + e^{-3t}(\frac{24}{25}\cos(4t) + \frac{147}{100}\sin(4t))\big)u(t)$
4 $h(t) = \big(2e^{-4t} + e^{-3t}(1 - 2t + 3t^2)\big)u(t)$
5(a) $H(s) = \dfrac{1}{s^2 + 2s + 2}$
5(b) $y(t) = e^{-t}\big(1 - \cos t\big)u(t)$
6(a) $y_{\text{zi}}(t) = 2e^{-2t}(2 + t)u(t)$
6(b) $y_{\text{zs}}(t) = \big(2 + 3te^{-2t}\big)u(t)$
6(c) $y_{\text{tot}}(t) = \big(2 + (4 + 5t)e^{-2t}\big)u(t)$
7 $h(t) = 2e^{-t}(1 - t)u(t)$
8(a) $H(s) = \dfrac{s}{(s+3)^2}$
8(b) $y(t) = te^{-3t}u(t)$
9(a) $H(s) = \dfrac{s^2 + 2}{(s+1)(s+2)}$
9(b) Notch filter
9(c) $h(t) = \delta(t) + (3e^{-t} - 6e^{-2t})u(t)$
9(d) $y(t) = (1 - 3e^{-t} + 3e^{-2t})u(t)$